

Page 1 — Machine Learning Fundamentals

Knowledge Base Document: Machine Learning | Page 1 of 5

Machine learning is a field in which algorithms learn patterns from data to make predictions or decisions. Supervised learning uses labeled examples, unsupervised learning discovers structure without target labels, and reinforcement learning learns through interaction and reward.

In supervised learning, regression predicts continuous values while classification predicts discrete classes. Examples include house-price prediction, customer churn classification, and credit-default prediction.

A dataset is commonly divided into training, validation, and test sets. The training set is used to fit model parameters, validation data supports model selection or tuning, and the test set provides an unbiased final estimate when used correctly.

Features are input variables and the target is the quantity being predicted. Data leakage occurs when information unavailable at prediction time enters the features or preprocessing process, producing misleadingly strong results.

Generalization is the ability of a model to perform well on unseen data. Underfitting occurs when a model is too simple, while overfitting occurs when a model learns noise or peculiarities of the training data.

Page 2 — Data Preparation and Feature Engineering

Knowledge Base Document: Machine Learning | Page 2 of 5

Data preparation includes handling missing values, inconsistent categories, duplicate records, invalid values, and inappropriate data types. The correct treatment depends on the meaning and distribution of the data.

Numerical features may require scaling for algorithms that depend on distances or gradient optimization. Standardization transforms values using mean and standard deviation, while min-max scaling maps values to a bounded interval.

Categorical variables can be represented using one-hot encoding, ordinal encoding, target encoding, or learned embeddings depending on the model and task. The encoding strategy should respect the semantics of the categories.

Feature engineering creates informative variables from raw data. Examples include ratios, date-derived features, aggregation over time windows, interaction terms, and domain-specific transformations.

Preprocessing should be fitted only on training data and then applied to validation and test data. Scikit-learn Pipelines and ColumnTransformer help enforce this discipline and reduce leakage.

Page 3 — Common Machine Learning Algorithms

Knowledge Base Document: Machine Learning | Page 3 of 5

Linear regression models a continuous target as a weighted combination of features. Logistic regression models class probabilities using a logistic function and is a strong baseline for binary classification.

Decision trees recursively split observations according to feature conditions. Random forests combine many trees using bagging and randomized feature selection. They are generally robust and can model nonlinear relationships.

Gradient boosting builds an ensemble sequentially, where later learners focus on errors made by previous learners. XGBoost and LightGBM are popular implementations designed for efficient boosting on structured data.

K-nearest neighbors predicts using nearby training examples. Support Vector Machines construct decision boundaries with maximum-margin principles and can use kernels for nonlinear relationships.

Algorithm selection should consider data size, feature types, interpretability requirements, latency, robustness, and maintenance cost. A slightly less accurate model can be preferable when it is easier to explain and operate.

Page 4 — Model Evaluation and Tuning

Knowledge Base Document: Machine Learning | Page 4 of 5

For classification, accuracy measures the fraction of correct predictions, while precision measures the fraction of predicted positives that are actually positive. Recall measures the fraction of actual positives detected. F1 is the harmonic mean of precision and recall.

ROC-AUC measures ranking quality across classification thresholds. Precision-recall curves can be more informative for highly imbalanced positive classes. The decision threshold should be selected according to business costs rather than automatically fixed at 0.5.

For regression, common metrics include MAE, MSE, RMSE, and R-squared. MAE is easier to interpret in target units, while MSE and RMSE penalize large errors more strongly.

Cross-validation estimates performance across multiple train-validation splits. Hyperparameter search can use GridSearchCV or RandomizedSearchCV. Search spaces should be realistic; excessively broad searches increase computation without guaranteeing useful improvement.

Evaluation must reflect the deployment scenario. Time-dependent data often requires chronological validation. Grouped observations may require GroupKFold. Metrics should be calculated on untouched evaluation data.

Page 5 — Production Machine Learning

Knowledge Base Document: Machine Learning | Page 5 of 5

A production ML system includes data ingestion, validation, preprocessing, model inference, monitoring, and feedback. Model training code should be reproducible and versioned separately from ad-hoc experimentation.

Model serialization allows a trained model to be loaded for inference. The serialized artifact should be compatible with the serving environment and treated as executable code when applicable; untrusted model files should not be loaded blindly.

Monitoring should track data quality, feature distributions, prediction distributions, latency, errors, and business outcomes. Concept drift occurs when the relationship between inputs and targets changes over time.

Explainability techniques can help identify important features and investigate individual predictions. Feature importance from tree models and SHAP-style explanations answer related but not identical questions and should be interpreted carefully.

Responsible ML requires attention to privacy, security, fairness, robustness, and human oversight. A model is not production-ready merely because its offline metric is high.